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GATE EC Question Analysis using Seven Segment Display

Question

If $X = 1$ in the logic expression

$$[X + Z(\bar{Y} + (\bar{Z} + X\bar{Y}))][\bar{X} + \bar{Z}(X + Y)] = 1$$

Find the condition for output to be 1.

Question Analysis:

Substitute $X = 1$

$$[1 + Z(\bar{Y} + (\bar{Z} + \bar{Y}))][0 + \bar{Z}]$$

$$= 1 \cdot \bar{Z}$$

$$\therefore F = \bar{Z}$$

The Truth Table

Z	F
0	1
1	0

Hardware Implementation

The above problem is implemented and tested in hardware using Arduino UNO board. A seven segment display is used to display output F as decimal digit 0 or 1 based on input Z. The Arduino reads input Z, computes $F = \bar{Z}$ and displays the output.

Required Components & Pin Connections

S.No	Component
1	Arduino Uno Board
2	Breadboard
3	Seven segment display
4	Resistors: 220Ω
5	Jumper Wires
6	USB Cable

Component	Arduino Pin
Input Z	Digital 9
Segment a	Digital 2
Segment b	Digital 3
Segment c	Digital 4
Segment d	Digital 5
Segment e	Digital 6
Segment f	Digital 7
Segment g	Digital 8
GND	GND
VCC	5V

Logic Description

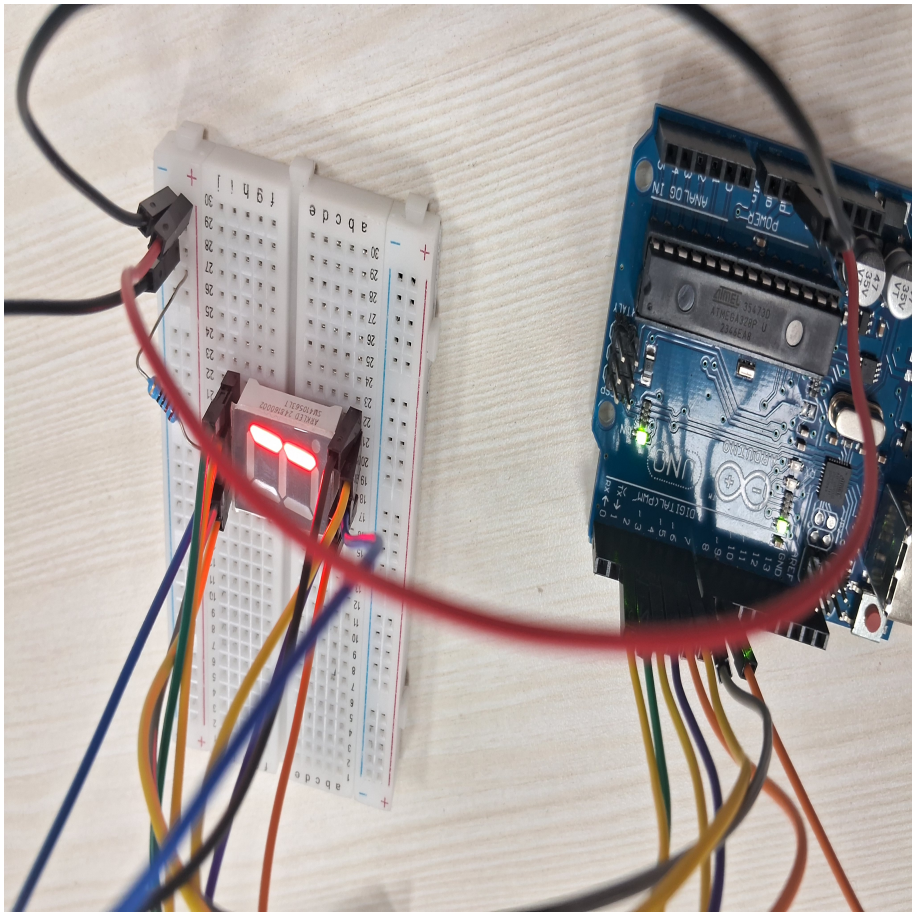
- Read input Z
- Simplified Boolean expression: $F = \bar{Z}$
- If $Z = 0$ display digit 1
- If $Z = 1$ display digit 0

Code Uploading Steps

1. Create Platform IO project
2. Write code in main.cpp
3. Compile using command "pio run"
4. Connect Arduino UNO using USB
5. Upload compiled code
6. Observe output on seven segment display

Experimental Truth Table

Z	Display Output
0	1
1	0



Conclusion

- The Boolean expression simplifies to $F = \bar{Z}$
- Hardware output matches theoretical truth table
- Seven segment display successfully verifies the logic